

A VR Studio for Media Art and Performance

Dave Pape*
University at Buffalo

Josephine Anstey
University at Buffalo

ABSTRACT

We discuss our experiences running a Virtual Reality studio in the context of a media studies department, supporting art and performance projects.

Keywords: Virtual reality, art, theatre.

Index Terms: I.3.7 [Three-Dimensional Graphics and Realism]: Virtual reality; J.5 [Arts and Humanities]: Performing arts

1 INTRODUCTION

Immersive visualization laboratories within large institutions are most often associated with research in science, engineering, and medicine. However, this does not always need to be the case; as has become clear in recent years, virtual reality is also a medium with great potential in the arts. Probably the most prominent early example of this is the VR program in the early 1990s at the Banff Centre [9]. In this paper, we will describe our own work running a VR lab primarily supporting the media arts and live performance.

The University at Buffalo's "Virtual Reality Studio" was established in the Department of Media Study (DMS) in the year 2000. DMS has a long history as an internationally known program in experimental and independent media [16]. From its very beginnings in 1973, the department has investigated not only film and video, but also digital arts, and it was natural to want to expand the research and curriculum to include VR.

2 SETTING UP THE STUDIO

We began the VR Studio as an offshoot of our previous work with the CAVE at the University of Illinois at Chicago [3]. As the Department of Media Study is an arts, rather than STEM research, program, funding (and space) for a VR system was much more limited. Fortunately, this was at a time that commodity off-the-shelf hardware and software were beginning to approach the capabilities of the high-end systems that were the basis of the original CAVE. Other researchers were starting to explore the use of game engines on PCs to drive CAVE-like displays [8], and we ourselves had experimented with newer, cheaper displays such as consumer flat panels for an ImmersaDesk-like system [10].

The VR Studio's initial system consisted of a Linux PC with dual-head graphics card (uncommon at the time), two LCD projectors, circularly polarizing filters and glasses, and a smaller (roughly 6' x 4') rear-projection screen [12]. When funding allowed, about two years later, we upgraded to a 12' x 8' screen. The remaining major component necessary for VR, which did not exist in mass-market form, was a 6 DOF tracker. The Ascension Technology SpacePad system, previously used with ImmersaDesks [4], however, was affordable for our purposes.

* email: depape@buffalo.edu

3 PROJECTS

The Virtual Reality Studio has been part of numerous projects over the years, some that would be classified as "traditional" VR work, but also several that have adapted the components of our system in different ways. These projects have included interactive narratives, installation art, theatrical performances, and scientific visualization.

3.1 Visualization

Visualization work included a collaboration with UB's Center for Computational Research and the Hauptmann-Woodward Institute, developing collaborative VR interfaces for analyzing protein structures via computational X-ray crystallography [6]. One part of this project included brief experiments with using a game engine as a visualization platform, although the software of the time was not fully up to the demands of this project. Another project, with the School of Medicine, explored synchronization of physical and virtual props for a better haptic interface in surgical training [5].

3.2 VR narrative and installations

Our initial media art projects were a continuation of past work in interactive, first-person narrative in VR. This grew into a collaboration with AI researchers from Computer Science and performance theorists from the Department of Theatre and Dance. "The Trial The Trail" (figure 1), and its successor "Human Trials", were networked, participatory dramas where a single audience member entered a narrative world featuring characters, some of which were controlled by human actors using head-mounted displays and others by AI agents [1]. These performances were presented in local art galleries, requiring the VR system to be portable, rather than fixed in the studio.



Figure 1. The Trial The Trail

Later projects continued to use the portability of our system to present work in a variety of settings, two examples being "PAAPAB" and "Art Flying In and Out of Space". "PAAPAB" was a distributed virtual disco, where participants could dance in

real time with others at remote locations, or record the motion-tracking data of their dances to be saved and then played back in an endless loop on the dance floor. This project was developed in collaboration with the Electronic Visualization Laboratory (Chicago) and the SARA supercomputing center (Amsterdam). It was also exhibited in non-networked form at a fundraising party for Hallwalls, a Buffalo art gallery. “Art Flying In and Out of Space” was a project led by artist Jackie Matisse, to create VR versions of her physical kite art; it was exhibited using the Studio’s VR system in the Zone Chelsea gallery in Manhattan [15].

3.3 Technological theatre

More explicitly theatre-centered work took the facilities of the VR Studio and remixed them in new ways appropriate to the needs of the projects. “365 Days / 365 Plays” featured a set of nine short plays by Suzan-Lori Parks (part of a coordinated national project between many groups) [14]. In staging some of the plays, we used stereoscopic projections for virtual sets. In another, the play was performed solely by AI agents. And in one play (“Diva”), the 6 DOF tracking system was used by an actor on stage, for real-time animation of a giant projected alter-ego (figure 2). Some of these technical elements were subsequently re-adapted to create the “Office Diva” project, a motion-captured virtual agent, who was presented as an installation in an abandoned office space, on a projection screen of old line-printer paper [11].



Figure 2. The Diva, from “365 Days / 365 Plays”

“WoyUbu” was a larger scale theatrical project, involving some of our regular collaborators and the experimental performance group the Real Dream Cabaret [13]. “WoyUbu” was a mashup of the plays *Woyzeck* (by Georg Buchner) and *Ubu Roi* (by Alfred Jarry). It made use of projections, video games, animation, performing robots, and surveillance technology. Projections provided virtual sets, in one performance area via chromakeying and in another via a 30-foot-long Powerwall-style multi-projector display of computer animation (figure 3). Battle scenes were performed by AI-controlled cleaning robots. The final scene used motion tracking in the *Ubu* performers’ area to destroy the virtual set of the *Woyzeck* performers.



Figure 3. WoyUbu

3.4 Students and visitors

In addition to the above projects, the VR Studio has served as a facility to support students and visitors to the university. Student MFA thesis projects have been created and exhibited in galleries using Studio systems, an example being Michael Chernoff’s recent “VIDEOSPHERE” project [2]. Visiting artists and researchers have relied on the Studio to present their work locally, such as PublicVR’s “The Egyptian Oracle” project [7].

4 CONCLUSION

We have given a brief overview of several of the projects that have been developed in the Virtual Reality Studio at the University at Buffalo. These include a range of performances and artworks involving many different specialists.

Over the years we have found that a projection-based system is highly amenable to the sorts of public-facing work that we have been involved with. As a PC-driven, single-screen system, it has been highly portable, not forcing events to happen within a campus lab, as would happen with a much larger device. Moreover, the low-cost, “homebrew” nature of the system has been an advantage rather than a detriment. The individual components – projectors, stereo imaging, 6 DOF tracking, virtual agent software, etc – have been highly flexible and readily adapted into many different applications. Although many of the specific elements that we started with two decades ago have become obsolete, we hope to continue with this paradigm, to supplement newer work with off-the-shelf, consumer HMD devices.

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